

BUSINESS REQUIREMENTS DOCUMENT

Banking Data Warehouse — ETL Testing and Data Quality Validation

Business Domain: Banking

Application Domain: Enterprise Data Warehouse / Regulatory Reporting

Testing Type: ETL Testing / Data Warehouse Testing / Data Quality Testing

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Confidentiality: Internal Use Only

1. Document Purpose

This Business Requirements Document defines the business requirements, data requirements, business rules, source systems, target systems, transformation requirements, data-quality expectations, reconciliation requirements, and validation requirements for the Bank's Enterprise Data Warehouse (EDW).

It establishes the baseline for ETL test planning, test-condition identification, scenario generation, source-to-target validation, business-rule validation, data-quality validation, regression testing, and production deployment validation.

2. Business Background

The Bank operates multiple transactional and operational systems supporting customer onboarding, deposits, loans, credit cards, payments, transactions, and account management. Operational data is generated across heterogeneous source systems and periodically consolidated into the Enterprise Data Warehouse.

The EDW provides information to Finance, Risk Management, Credit Risk, Compliance, AML, Regulatory Reporting, Customer Analytics, Management Reporting, Business Intelligence, Fraud Detection, and Audit functions.

3. Business Objectives

ID	Business Objective
BO-001	Ensure complete extraction of eligible records from source systems.
BO-002	Ensure accurate transformation according to approved business rules.
BO-003	Ensure all valid records are loaded into appropriate target structures.
BO-004	Prevent duplicate customer, account, transaction, and loan records.
BO-005	Ensure source-to-target data reconciliation.
BO-006	Validate data quality across critical banking data elements.
BO-007	Ensure incremental loads process only eligible records.
BO-008	Validate historical data retention and Slowly Changing Dimension behavior.
BO-009	Ensure ETL jobs complete within the defined batch window.
BO-010	Ensure downstream regulatory and management reports receive accurate data.
BO-011	Detect rejected, quarantined, and exception records.
BO-012	Provide auditability and traceability of data movement.

4. Scope

4.1 In Scope

Source-to-staging validation; staging-to-warehouse validation; source-to-target reconciliation; transformation and business-rule validation; full, incremental, initial and historical loads; Slowly Changing Dimensions; duplicate, null, referential-integrity, data-type, length, format and boundary validation; reject/error handling; batch controls; audit columns; dependency validation; restart/recovery; completeness, accuracy, consistency and timeliness; regression testing.

4.2 Out of Scope

Functional testing of source banking applications, UI testing, mobile/internet banking UI testing, customer-facing application performance testing, penetration testing, network infrastructure testing, and infrastructure disaster-recovery testing unless explicitly included.

5. Stakeholders

Stakeholder	Responsibility
Business Data Owner	Defines business meaning and acceptance criteria for critical data.
Business Analyst	Documents business requirements and business rules.
Data Architect	Defines target data architecture and data model.
ETL Developer	Develops extraction, transformation and loading processes.
ETL Test Engineer	Validates ETL processing and source-to-target data movement.
Data Quality Analyst	Validates data-quality dimensions and thresholds.
DBA	Supports database access, SQL execution and environment management.
Regulatory Reporting Team	Validates regulatory reporting requirements.
Risk Management	Validates risk-related data requirements.
Compliance Team	Validates compliance and AML-related data requirements.
Production Support	Supports batch execution and production incidents.
Release Manager	Coordinates deployment and release activities.

6. Source Systems

Source System	Description	Primary Data
CBS	Core Banking System	Customer, account, deposit and transaction data
CRM	Customer Relationship Management	Customer profile and relationship data
LOS	Loan Origination System	Loan application and origination data
LMS	Loan Management System	Loan account, repayment and outstanding balance data
CMS	Card Management System	Credit/debit card information and transactions
Payment Gateway	Payment processing platform	Payment transactions
AML System	Anti-Money Laundering system	Alerts, cases and monitoring data
Treasury System	Treasury operations	Investment and liquidity data
General Ledger	Financial accounting system	GL accounts and financial balances

7. Target Systems

The logical data flow is:

Source Systems → Landing Layer → Staging Layer → Operational Data Store → Enterprise Data Warehouse → Data Marts → Reporting / Analytics / Regulatory Systems

Target subject areas include Customer, Account, Deposit, Loan, Card, Transaction, Payment, Branch, Product, General Ledger, Customer Risk, and Regulatory Reporting.

8. Business Requirements

BR-001 — Customer Data Integration

The ETL process shall extract eligible customer records from approved source systems and load them into the Customer dimension. Relevant attributes include Customer ID, Customer Name, Date of Birth, Customer Type, Customer Segment, Customer Status, Customer Risk Rating, Residency Status, Country, Customer Onboarding Date, and KYC Status. Customer ID shall uniquely identify a customer within the enterprise customer model.

BR-002 — Account Data Integration

The ETL process shall extract account information from the Core Banking System and load eligible records into Account dimension/fact structures. Attributes include Account Number, Customer ID, Account Type, Product Code, Branch Code, Currency, Account Open Date, Account Close Date, Account Status, Available Balance, and Ledger Balance. Each account shall be associated with a valid customer.

BR-003 — Loan Data Integration

Loan data shall be extracted from LOS and LMS. The warehouse shall maintain Loan Account Number, Customer ID, Loan Product, Principal Amount, Interest Rate, Tenure, Disbursement Date, Maturity Date, Outstanding Principal, Outstanding Interest, Loan Status, Delinquency Status, and Days Past Due.

BR-004 — Transaction Data Integration

Eligible financial transactions shall be loaded into the transaction fact table. Attributes include Transaction ID, Account Number, Customer ID, Transaction Date, Transaction Type, Debit/Credit Indicator, Transaction Amount, Currency, Channel, Branch, Transaction Status, and Source System Identifier. Transaction ID shall support transaction-level reconciliation.

9. Business Rules

Rule ID	Business Rule
BRULE-001	A customer shall have only one active enterprise customer record for the same source business identifier. Duplicate records shall be rejected or quarantined according to the approved data-quality framework.
BRULE-002	Every active account shall be associated with a valid customer. An invalid Customer ID shall be rejected and recorded in the ETL exception mechanism.
BRULE-003	Transaction Amount shall be greater than zero for completed financial transactions unless the transaction type explicitly permits zero-value transactions.
BRULE-004	Debit/Credit Indicator shall contain only D = Debit or C = Credit.
BRULE-005	Account Status shall contain only ACTIVE, DORMANT, CLOSED, BLOCKED, or SUSPENDED.
BRULE-006	Days Past Due shall be calculated using Business Date minus Contractual Payment Due Date; for accounts with no overdue amount, Days Past Due shall be zero.
BRULE-007	Customer risk classification shall follow the approved risk-rating rules and retain the classification applicable to the reporting date.

10. Source-to-Target Transformation Requirements

Source	Transformation	Target
CUST_ID	Direct mapping	CUSTOMER_ID
CUST_NAME	Trim leading/trailing spaces	CUSTOMER_NAME
DOB	Convert to warehouse date format	DATE_OF_BIRTH
ACCT_NO	Direct mapping	ACCOUNT_NUMBER
BAL_AMT	Currency conversion where applicable	BALANCE_AMOUNT
TXN_DT	Date standardization	TRANSACTION_DATE
DR_CR_FLAG	Code mapping	DEBIT_CREDIT_IND
LOAN_AMT	Direct mapping	LOAN_PRINCIPAL_AMOUNT

The ETL test team shall validate every critical source-to-target mapping and transformation rule.

11. Data Quality Requirements

11.1 Completeness

All eligible source records shall be loaded unless explicitly rejected by an approved business rule. Eligible Source Records = Target Successfully Loaded Records + Rejected Records.

11.2 Accuracy

Target values shall accurately represent corresponding source values after applying approved transformations.

11.3 Consistency

The same business entity shall have consistent values across relevant warehouse tables and downstream data marts.

11.4 Uniqueness

Business keys such as Customer ID, Account Number and Transaction ID shall not contain unintended duplicates.

11.5 Validity

Values shall conform to approved formats, code sets, data types, ranges and reference data.

11.6 Timeliness

ETL processes shall complete within the defined SLA and downstream reporting shall not start until prerequisite ETL jobs complete successfully.

12. Incremental Load Requirements

The ETL framework shall support incremental processing using Last Updated Timestamp, CDC timestamp, source sequence number, CDC indicator, or batch control number as applicable. The solution shall process new, updated, deleted, unchanged, late-arriving and duplicate incremental records according to the approved design. Boundary timestamp records shall be explicitly tested.

13. Slowly Changing Dimension Requirements

Customer and other master/reference entities shall support historical tracking where required. For SCD Type 2, the warehouse shall maintain Customer ID, Surrogate Key, Customer Name, Customer Status, Effective From Date, Effective To Date, and Current Flag. When a tracked attribute changes, the existing record shall be expired, a new version created, current-flag values updated, effective dates maintained without overlap, and historical information preserved.

14. Reject and Exception Handling

Records failing validation shall be routed to an approved reject or exception mechanism. Reject information should include Batch ID, Source System, Source Record ID, Error Code, Error Description, Rejection Timestamp, ETL Job Name, and Processing Date. Invalid records shall not be silently discarded.

15. Reconciliation Requirements

Source-to-target reconciliation shall be performed for critical data entities. Any reconciliation difference shall be investigated, explained, and documented.

16. Audit Requirements

The warehouse shall maintain audit information such as Batch ID, ETL Job ID, Source System, Source File Name, Load Start Timestamp, Load End Timestamp, Created Timestamp, Updated Timestamp, Record Status, and Processing Date to support end-to-end lineage and operational troubleshooting.

Source Eligible Count = Target Successfully Loaded Count + Rejected Count

For financial transactions:

Source Transaction Amount = Target Transaction Amount + Approved Adjustment

17. Batch Processing Requirements

Customer Extract → Customer Staging → Customer Dimension → Account Load → Transaction Load → Data Mart Load → Regulatory Reporting

A dependent job shall not execute successfully when its prerequisite job has failed.

18. ETL Testing Requirements

Testing Category	Objective
Source Validation	Validate source data availability and structure.
Extraction Testing	Validate correct extraction of eligible records.
Transformation Testing	Validate transformations and business rules.
Load Testing	Validate target loading.
Source-to-Target Testing	Validate mapping correctness.
Data Completeness Testing	Validate all eligible records are processed.
Data Accuracy Testing	Validate target values.
Data Quality Testing	Validate quality dimensions.
Reconciliation Testing	Validate counts and financial amounts.
Duplicate Testing	Validate uniqueness.
Referential Integrity Testing	Validate relationships.
SCD Testing	Validate historical data.
Incremental Load Testing	Validate delta processing.
Negative Testing	Validate invalid-data handling.
Boundary Testing	Validate limits and edge conditions.
Exception Testing	Validate rejected records.
Restart/Recovery Testing	Validate batch recovery.
Regression Testing	Ensure existing functionality remains unaffected.
Performance Testing	Validate ETL execution within SLA.

19. Critical Banking Data Elements

The following shall be treated as Critical Data Elements (CDEs): Customer ID, Account Number, Transaction ID, Loan Account Number, Transaction Amount, Account Balance, Customer Risk Rating, Transaction Date, Account Status, Loan Status, Currency, Branch Code, Product Code, General Ledger Account, and Regulatory Classification.

20. Regulatory and Compliance Considerations

The ETL solution shall preserve data required for regulatory reporting, financial reporting, AML monitoring, KYC reporting, risk reporting, audit, data lineage, and historical traceability. Transformations shall not result in loss of regulatory-relevant information.

21. Security and Data Access Requirements

Sensitive banking information shall be protected according to organizational security policies. Sensitive data includes customer personal information, account information, transaction information, identification information, and financial information. Production data shall not be exposed to unauthorized users. Masking shall be applied in non-production environments where required.

22. Data Retention Requirements

Historical banking data shall be retained according to the Bank's approved data-retention policy, considering regulatory, audit, business, legal, and reporting requirements. Historical records shall not be unintentionally overwritten or deleted.

23. Performance and SLA Requirements

Process	Example SLA
Source Extraction	01:00 AM – 02:00 AM
Staging Load	02:00 AM – 02:45 AM
Warehouse Load	02:45 AM – 04:00 AM
Data Mart Load	04:00 AM – 04:30 AM
Reconciliation	04:30 AM – 04:45 AM
Reporting Availability	05:00 AM

The testing team shall validate that processing completes within the agreed production SLA under expected production volumes.

24. Error Handling Requirements

The ETL framework shall capture processing errors, generate appropriate error codes, maintain reject records, log job execution details, identify failed stages, preserve failed source records where applicable, support restartability, prevent duplicate processing after restart, and generate operational alerts for critical failures.

25. Restart and Recovery Requirements

If an ETL job fails, restart processing shall not cause duplicate records, duplicate transactions, partial financial balances, incorrect SCD versions, or inconsistent target data. Restart processing shall follow the approved checkpoint or recovery mechanism.

26. Acceptance Criteria

- 100% of critical source-to-target mappings are validated.
- All critical business rules are tested.
- All CDEs are validated.
- Source and target counts reconcile within approved tolerances.
- Financial amounts reconcile within approved tolerances.
- No unexplained duplicate CDE records exist.
- Referential integrity violations are within approved thresholds.
- Reject records are correctly captured.
- Incremental loads process eligible changes correctly.
- SCD behavior is validated.
- ETL jobs complete within the agreed SLA.
- Audit information is available.

- Critical and high-severity defects are resolved or formally accepted.
- Business stakeholders approve the test results.

27. Sample ETL Test Scenarios

Scenario 1 — Customer Record Completeness

Objective: Verify that all eligible customer records are loaded. Source: CBS Customer Table. Target: DIM_CUSTOMER.

Validation: Source Eligible Customer Count = Target Customer Count + Rejected Customer Count.

Scenario 2 — Account-Customer Referential Integrity

Objective: Verify every active account has a valid Customer ID.

Expected Result: 0 records without a matching Customer ID.

Scenario 3 — Transaction Amount Validation

Objective: Validate transaction amount transformation. Example source TXN_AMT = 12500.50; expected target TRANSACTION_AMOUNT = 12500.50.

Scenario 4 — Duplicate Transaction Validation

Objective: Ensure Transaction ID is unique. Expected Result: 0 duplicate Transaction IDs.

Scenario 5 — SCD Type 2 Validation

When Customer Status changes from ACTIVE to DORMANT, the ETL process shall expire the existing ACTIVE record, create a new DORMANT version, update CURRENT_FLAG appropriately, maintain effective dates, and preserve historical data.

Scenario 6 — Incremental Load Validation

Given Last Successful Batch Timestamp = T1, records satisfying LAST_UPDATED_TIMESTAMP > T1 shall be processed as eligible incremental changes, subject to the approved CDC design.

28. Business Rules to Test

Rule ID	Rule
BRULE-001	Customer ID must be unique.
BRULE-002	Every active account must have a valid Customer ID.
BRULE-003	Completed financial transactions must have valid transaction amounts.
BRULE-004	Debit/Credit Indicator must contain approved values.
BRULE-005	Account Status must contain approved values.
BRULE-006	Days Past Due must use approved business-date logic.
BRULE-007	Customer risk classification must follow approved risk rules.
BRULE-008	Transaction ID must be unique.
BRULE-009	Currency must be a valid ISO currency code.
BRULE-010	Branch Code must exist in approved branch reference data.

29. Requirements Traceability

Business Requirement → Business Rule → Source System → Source Attribute → Transformation Rule → Target Attribute → Test Scenario → Test Case → Test Execution → Defect → Test Evidence

Every requirement shall be traceable through the testing lifecycle. The traceability matrix shall establish coverage between business requirements and ETL validation.

30. Requirement Identification Structure

Each requirement shall have a unique identifier such as REQ-CUST-001, REQ-ACCT-001, REQ-LOAN-001, REQ-TXN-001, REQ-DQ-001, REQ-RECON-001, REQ-SCD-001, and REQ-ETL-001.

Each requirement should contain Requirement ID, Requirement Description, Requirement Type, Business Domain, Source System, Source Table, Source Column, Target System, Target Table, Target Column, Business Rule, Transformation Rule, Data Quality Rule, Priority, Criticality, and Acceptance Criteria.

31. Expected ETL Test Deliverables

- ETL Test Strategy
- ETL Test Plan
- Source-to-Target Validation Matrix
- Business Rule Validation Matrix
- Data Quality Rules
- Test Scenarios
- Test Cases
- SQL Validation Scripts
- Reconciliation Reports
- Defect Reports
- Test Execution Reports
- Requirement Traceability Matrix
- Test Summary Report
- Production Validation Checklist

32. Entry Criteria

- BRD is approved.
- Source-to-target mapping is available.
- ETL design is approved.
- Business rules are documented.
- Test environment is available.
- Required source data is available.
- Target tables are deployed.
- Test data is available.
- ETL jobs are deployed.
- Reference data is available.

33. Exit Criteria

- All planned test scenarios are executed.
- All critical business rules are validated.
- All CDEs are validated.
- Reconciliation is completed.
- No open Critical defects remain.
- No open High-severity defects remain unless formally accepted.
- Regression testing is completed.

- Test evidence is available.
- Business stakeholders approve test results.

34. Assumptions

- Source systems provide accurate transactional data.
- Source-to-target mappings are approved by the Data Architect and Business Data Owner.
- Reference data is available before ETL processing.
- Required database access is available.
- Batch schedules are established before integration testing.
- Business rules are approved before test execution.
- Data retention requirements are defined by the appropriate governance team.

35. Dependencies

- Source system availability
- ETL development completion
- Source-to-target mapping availability
- Test environment availability
- Reference data availability
- Database connectivity
- Batch scheduler availability
- Test data availability
- Business rule approval
- Data warehouse deployment

36. Risks

Risk	Impact	Mitigation
Source data unavailable	Testing delay	Establish source-data availability SLA.
Incorrect mapping	Incorrect target data	Perform mapping review.
Missing business rules	Incomplete coverage	Conduct business walkthrough.
Large data volumes	Performance degradation	Conduct volume testing.
Late-arriving data	Reporting inconsistency	Implement approved late-arriving-data handling.
Duplicate source records	Data-quality issues	Apply duplicate validation.
Reference data mismatch	Invalid target records	Validate reference data.
Batch failure	Reporting delay	Implement restart/recovery mechanism.

37. Business Acceptance

The ETL solution shall be accepted when the Business Data Owner, Data Quality Team, ETL Testing Team, and relevant downstream stakeholders confirm that the implemented solution satisfies the approved business requirements and data-quality expectations.

Approval Role	Signature	Date
Business Data Owner		
ETL Test Lead		
Data Architect		
Business Analyst		
Project Manager		

Appendix A — Banking Domain Terminology

Core Banking System (CBS); Customer Information File (CIF); Customer ID; Account Number; General Ledger (GL); Loan Account; Credit Facility; Deposit Account; Transaction; Debit/Credit Indicator; Branch Code; Product Code; Customer Segment; Customer Risk Rating; KYC; AML; Regulatory Reporting; Data Warehouse; Enterprise Data Warehouse (EDW); Data Mart; Landing Layer; Staging Layer; Operational Data Store (ODS); Fact Table; Dimension Table; Surrogate Key; Natural Key; Business Key; Slowly Changing Dimension (SCD); Source-to-Target Mapping (STTM); Change Data Capture (CDC); Full Load; Incremental Load; Batch Processing; Batch ID; Reconciliation; Data Lineage; Critical Data Element (CDE); Data Quality; Referential Integrity; Exception Handling; Reject Processing; Audit Trail; Effective Date; Expiry Date; Current Flag; Business Date; Cut-off Time; Processing Date; Reporting Date; SLA; Data Retention; Regulatory Compliance.

Appendix B — Recommended Test Coverage Model

ETL TESTING

DATA FLOW | BUSINESS LOGIC | DATA QUALITY

Source → Target | Transformations | Completeness

Extraction | Business Rules | Accuracy

Loading | Derivations | Consistency

Reconciliation | Calculations | Uniqueness

| | Validity | Timeliness

Cross-cutting: Security | Auditability | Data Lineage | Exception Handling | Restartability |

Performance | Regression | Regulatory Compliance

Appendix C — Primary Requirement-to-Scenario Relationship

Requirement:

Every active account must have a valid Customer ID.

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Business Rule:

ACCOUNT.CUSTOMER_ID must exist in CUSTOMER.CUSTOMER_ID.

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Test Scenario:

Validate referential integrity between Account and Customer.

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Positive Scenario:

Active account contains a valid Customer ID.

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Negative Scenario:

Active account contains an invalid Customer ID.



Boundary / Exception Scenario:

Customer record arrives after Account record.



Validation:

Target Account record must be associated with the customer or handled according to the approved late-arriving-dimension strategy.
